

Experiment – 1 a: TypeScript

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Experiment – 1 a: TypeScript

Aim: Write a simple TypeScript program using basic data types (number, string, boolean) and operators.

Description :

Create a calculator in TypeScript that uses basic operations like addition, subtraction, multiplication, and division. It also gracefully handles invalid operations and division by zero..Design a Student Result database management system using TypeScript.

OUTPUT :

a) Calculator

```
C:\Users\Student.DESKTOP-HH2106Q\Desktop\folder>tsc Calculator.ts
C:\Users\Student.DESKTOP-HH2106Q\Desktop\folder>node Calculator.js
Enter the first number: 32
Enter the second number: 22
Enter operation (add, subtract, multiply, divide): add
Result: 54
```

```
C:\Users\Student.DESKTOP-HH2106Q\Desktop\folder>node Calculator.js
Enter the first number: 33
Enter the second number: 44
Enter operation (add, subtract, multiply, divide): add
Result: 77

C:\Users\Student.DESKTOP-HH2106Q\Desktop\folder>node Calculator.js
Enter the first number: 22
Enter the second number: 0
Enter operation (add, subtract, multiply, divide): divide
Result: Error: Division by zero is not allowed.

C:\Users\Student.DESKTOP-HH2106Q\Desktop\folder>node Calculator.js
Enter the first number: 33
Enter the second number: 8
Enter operation (add, subtract, multiply, divide): subtract
Result: 25

C:\Users\Student.DESKTOP-HH2106Q\Desktop\folder>node Calculator.js
Enter the first number: 8
Enter the second number: 5
Enter operation (add, subtract, multiply, divide): multiply
Result: 40
```

```
C:\Users\Student.DESKTOP-HH2106Q\Desktop\folder>node Calculator.js
Enter the first value: Anushka
Enter the second value: Shahane
Enter operation (add, subtract, multiply, divide): add
Result: Anushka Shahane

C:\Users\Student.DESKTOP-HH2106Q\Desktop\folder>node Calculator.js
Enter the first value: 44
Enter the second value: 2
Enter operation (add, subtract, multiply, divide): square
Error: Invalid operation.
```

OUTPUT :

B] Design a Student Result database management system using TypeScript.

```
C:\Users\Student.DESKTOP-HH2106Q\Desktop\folder>tsc student.ts

C:\Users\Student.DESKTOP-HH2106Q\Desktop\folder>node student.js
Student Name: Anushka Shahane
Average Marks: 44.333333333333336
Result: Passed
```

Github link : https://github.com/Anushka3204/WebX_Experiment_1

Conclusion : In this experiment, I successfully explored the foundational features of TypeScript, including data types, type annotations, and object-oriented programming concepts. By implementing a calculator using basic operations and handling errors such as invalid operations and division by zero, we understood how TypeScript ensures type safety and improves code readability and maintainability.